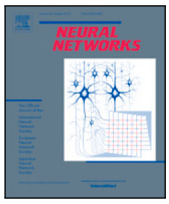
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Full Length Article



Large Language Model Enhanced Logic Tensor Network for Stance Detection

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ABSTRACT

Social media platforms, rich in user-generated content, offer a unique perspective on public opinion, making stance detection an essential task in opinion mining. However, traditional deep neural networks for stance detection often suffer from limitations, including the requirement for large amounts of labeled data, uninterpretability of prediction results, and difficulty in incorporating human intentions and domain knowledge. This paper introduces the First-Order Logic Aggregated Reasoning framework (FOLAR), an innovative approach that integrates first-order logic (FOL) with large language models (LLMs) to enhance the interpretability and efficacy of stance detection. FOLAR comprises three key components: a Knowledge Elicitation module that generates FOL rules using a chain-of-thought prompting method, a Logic Tensor Network (LTN) that encodes these rules for stance detection, and a Multi-Decision Fusion mechanism that aggregates LTNs' outputs to minimize biases and improve robustness. Our experiments on standard benchmarks demonstrate the effectiveness of FOLAR, showing it as a promising solution for explainable and accurate stance detection. The source code will be made publicly available to foster further research.

1. Introduction

Social media has emerged as a prevalent platform for users to voice perspectives on controversial topics. Aggregating and analyzing these perspectives enables identifying overarching trends and emotions on contentious issues spanning abortion to epidemic prevention (Zeng, Zhou, & Huang, 2023). This abundance of data presents immense potential for web mining and content analysis (AlDayel & Magdy, 2021). The insights obtained can be integral to diverse decision-making, from advertising to political campaigns. Therefore, automatic stance detection on social media has become imperative in opinion mining, allowing more nuanced comprehension of user attitudes on multifarious matters (Zhang, Fu, Luo et al., 2023).

Stance detection, a core opinion mining task, aims to categorize attitudes towards specific targets based on opinionated text (Wang, Sun, Li, Zhu, & Zhou, 2020). Current approaches typically construct text classification models to predict stance towards a target, which can be divided into non-pretrained or pretrained language models (PLMs). Non-pretrained models utilize deep neural networks (DNN) for stance classification, including long short-term memory (LSTM), attention-based methods, and graph convolutional networks (GCN). For instance, Du, Xu, He, and Gui (2017b) employed attention to leverage target-specific knowledge. Dey, Shrivastava, and Kaushik (2018) used two RNNs to filter and classify text separately. Sun, Wang, Zhu, and Zhou (2018) developed hierarchical attention to model representations through linguistic factors. Liang, Chen et al. (2022) presented an effective GCN method differentiating target-invariant and -specific features. Recently, fine-tuning and prompt-tuning PLMs have advanced performance (Liu, Lin, Tan, & Wang, 2021). Fine-tuning adapts PLMs via a stance classification head and model fine-tuning. Prompt-tuning reformulates tasks as cloze questions, achieving success in text classification (Schick & Schütze, 2021). A typical prompt-tuning approach packs input into templates for masked language modeling.

Nevertheless, end-to-end deep neural networks are often treated as "black boxes", posing limitations such as the need for extensive labeled data, lack of interpretability, and challenges in integrating human expertise and domain knowledge (Liu, Chatterjee, Zhou, Lu, & Abusorrah, 2020). To the best of our knowledge, research into the interpretability of stance detection remains sparse. Recent endeavors in the broader natural language processing domain related to stance detection have concentrated on constructing models founded on first-order logic (FOL) to enhance the explainability of neural classifiers (Bianchi

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& Hitzler, 2019; Hu, Ma, Liu, Hovy and Xing, 2016; Huang et al., 2022; Karthikeyan, McDonald, & Mehta, 2023). The focus on FOL-based approaches is attributed to their provision of a flexible declarative language that is advantageous for elucidating complex cognitive processes and encapsulating structured knowledge (Zhang, Huang et al., 2022).

While these FOL-based works provide inspiration for interpretable network construction, a persistent challenge remains in the need to manually define domain knowledge for specific targets to construct first-order logic rules for guiding model building. Specifying domain knowledge for task objectives is non-trivial, especially under zero-shot detection settings approaching real-world scenarios, thus constraining model applicability.

Recently, large language models (LLMs) have demonstrated remarkable capabilities across diverse applications. The semantic capabilities of LLMs provide a promising basis for formulating FOL definitions by leveraging their robust understanding of natural language semantics. Most LLMs can readily comprehend logical reasoning processes through zero-shot prompting, greatly enhancing the applicability of FOL-based interpretable models. Despite their potential to revolutionize the construction of stance detection models grounded in FOL, LLMs often exhibit bias in knowledge elicitation due to variations in their self-learning from training datasets and methodologies.

To effectively address the aforementioned deficiencies, in this paper, we propose the First-Order Logic Aggregated Reasoning framework (FOLAR) for stance detection. This framework automatically generates FOLs using a prompt learning approach and employs a logic tensor network to create FOL-based deep learning models for stance detection in an end-to-end manner. FOLAR consists of three components: knowledge elicitation module (KE), logic tensor network (LTN), and multi-decision fusion (MDF). The KE employs a chain-of-thought (CoT) prompting method grounded in first-order logic rules. We design prompt templates specifically to extract logical rules from LLMs, facilitating the elicitation of inference patterns that underpin the framework. The LTN harnesses the power of sample learning by encoding FOL into its structure, utilizing annotated data to its full potential. This alignment with first-order logic ensures that the model's decisions are both interpretable and data-driven. The MDF aims to reduce the bias from individual LLMs by developing an attention-based dynamic routing strategy that effectively aggregates LTNs built on inference rules from different models. This enhances the model's robustness and improves its accuracy, providing a more reliable stance detection process.

This work significantly extends our previous work SILTN (Zhang, Huang et al., 2022). This paper presents several key extensions: Firstly, the definition of FOL in SILTN required manual crafting, limiting its generalizability and accuracy. In contrast, we propose the KE method in this paper, which extracts FOLs from LLMs through a CoT approach, reducing the reliance on human expertise and enhancing model generalizability. Secondly, unlike the fixed network structure used in our previous work (Zhang, Huang et al., 2022), the FOLAR method allows for flexible modeling of FOL outputs from large models into LTN. The MDF module then integrates different LTNs using an attention algorithm to predict the final stance. Furthermore, we expanded the experiments of SILTN by introducing more advanced baseline methods to compare with our model, demonstrating the effectiveness of the FOLAR method. In this paper, some highly relevant sections from (Zhang, Huang et al., 2022) are omitted to emphasize our new contributions.

The main contributions of this paper are summarized as follows:

- To the best of our knowledge, this is the first attempt to integrate an interpretable logic tensor network into a stance detection framework. The FOLAR model, grounded in FOL, enhances the ability to articulate complex cognitive processes and encapsulate structured knowledge, marking a significant advancement in the field of explainable stance detection.
- We introduce the LTN, which can encode FOL via backpropagation, leveraging annotated data while preserving interpretability. Additionally, our MDF uses attention-based aggregation to reduce biases and enhance the accuracy and robustness of stance detection.
- Comprehensive experiments were conducted on commonly adopted benchmarks to validate the performance of FOLAR. The results demonstrate and affirm its effectiveness in stance detection. To promote further research and exploration, we will publicly release the source code of our model.

The remainder of this paper is structured as follows. We provide a comprehensive review of traditional and recent stance detection approaches in Section 2. Section 3 gives the details of our proposed FOLAR framework. The experimental results and analysis are presented in Section 4. Finally, we conclude the paper by summarizing the key findings and outlining potential avenues for future research in Section 5.

2. Related work

2.1. Stance detection

Stance detection aims to infer a text's attitude toward a specific target, intersecting with argument mining, fact-checking, and aspect-level sentiment analysis (Rani & Kumar, 2022; Zhou et al., 2024).

Traditional stance detection often utilizes machine learning models such as support vector machines (Patra, Das, & Bandyopadhyay, 2016), naive Bayes (Tan, Cheng, Wang, & Xu, 2009), logistic regression (Genkin, Lewis, & Madigan, 2007), etc. These models are well-defined and structurally simple. For example, a two-stage stance detection system based on SVM was proposed to analyze stances on Twitter (Dey, Shrivastava, & Kaushik, 2017). Mourad et al. demonstrated that random forests, linear SVM, and Gaussian naive Bayes can serve as majority vote classifiers for stance identification (Mourad, Shawky, Fayed, & Badawi, 2018). However, these models are linguistically demanding and susceptible to human error. As information volume grows, traditional stance detection models become more time-consuming and costly, making mining stance information increasingly challenging in the information age.

With the advancement of deep learning, DNNs have become the primary framework for stance detection. Current approaches predominantly rely on either non-pretrained or pretrained models (PLM-based models). Non-pretrained methods commonly employ deep neural networks including Attention and GCN for stance classification training. Attention-based models utilize target-specific data as the attention query to infer stance (Sun et al., 2018). GCNs introduce graph convolutions to model target-text relations (Cignarella, Bosco, & Rosso, 2022). Additionally, some studies have explored the integration of background knowledge to enhance the efficacy of stance detection, yielding significant improvements in both cross-target and zero-shot settings. For example, Liu et al. (2021) proposed a knowledge-enhanced graph model that leverages BERT to capture both inter- and extra-semantic information. Zhang, Yang et al. (2020) introduced an emotion lexicon-based approach to augment stance detection. Liang, Chen et al. (2022) developed a framework to distinguish between target-specific and invariant features, thereby improving transferable learning.

The PLM-based models focus on the application of fine-tuning and prompt-tuning framework in stance detection. Specifically, Jiang, Gao, Shen, and Cheng (2022) proposed a prompt-tuning approach tailored to stance detection. This framework leverages a verbalizer function to map each label to a hidden vector, thereby facilitating label prediction. Then, Hardalov, Arora, Nakov, and Augenstein (2022) proposed a prompt-tuning framework specifically designed for cross-lingual stance detection. However, these approaches predominantly treat the model as an end-to-end black box, resulting in the aforementioned challenges.

2.2. Chain-of-thought methods

Recent studies have explored enhancing CoT prompting to elicit impressive multi-hop reasoning from LLMs. For instance, Cai, Chang, and Han (2023) proposed a human-in-the-loop system augmented with CoT prompting, investigating how manual correction of sub-logic in rationales can refine LLM reasoning. Fei et al. (2023) introduced a multi-step CoT approach that decomposes downstream tasks into multiple stages to improve prediction effectiveness. Ling et al. (2023) presented a new CoT technique, which iteratively infers tasks via deductive reasoning and verification.

More recently, several approaches have been developed to construct stance detection technologies based on CoT, which LLMs to facilitate prediction. One category of methods utilizes LLMs as predictors, where zero-shot or one-shot prompt templates are constructed to enable direct stance prediction. For instance, Zhang, Ding and Jing (2022) first proposed a zero-shot prompt method, for LLM-based stance detection. In a subsequent study, Zhang, Fu, Ding et al. (2023) introduced a two-stage prompting method for stance detection, which utilizes a single case sample to construct the one-shot prompt. Another category of methods treats LLMs as knowledge repositories, where templates are employed to extract relevant background knowledge for stance detection (Ding et al., 2024). For example, Li, Liang et al. (2023) leverage LLMs to retrieve target-related background knowledge, thereby enhancing the accuracy of stance detection. Inspired by multi-step CoT techniques, we propose a novel knowledge elicitation method that effectively generates prediction rationales in FOL format.

2.3. Methods with first-order logic

Logic rules provide a flexible declarative language for communicating high-level cognition and expressing structured knowledge. Recently, several works have been developed to integrate FOLs into DNNs. For example, Hu, Ma et al. (2016) proposed a general distillation framework (DNNLR) that transfers knowledge expressed as first-order logic rules into neural networks, where FOL constraints are integrated via posterior regularization. Subsequently, Hu, Yang, Salakhutdinov and Xing (2016) extended DNNLR by integrating various knowledge. Zhang et al. (2019) further extended the training strategy based on posterior regularization. Generally, these methods expressed knowledge into FOL constraints integrated via posterior regularization.

In recent years, LTN methods have garnered widespread attention. LTN constitutes a neurosymbolic formalism and computational model, facilitating learning and reasoning by introducing a many-valued, end-to-end differentiable FOL for deep learning applications. LTN has been successfully employed in a wide range of tasks, encompassing perception, learning, and reasoning about relational knowledge. For example, LTN was applied to semantic image interpretation whereby relational knowledge about objects was injected into deep networks for object relationship detection (Badreddine, d'Avila Garcez, Serafini, & Spranger, 2022). Bianchi and Hitzler (2019) performed LTN for reasoning about ontological knowledge. Then Bianchi, Palmonari, Hitzler, and Serafini (2019) developed LTN for learning the embedding of concepts into a latent real space by taking into consideration ontological knowledge about such concepts. In summary, LTN supports end-to-end differentiable first-order logic as the representation language of deep learning, which makes it highly interpretable.

The semantics of LTN, referred to as real logic, deviate from the standard abstract semantics of FOL. Specifically, in real logic, every object denoted by constants, variables, and terms is interpreted as a tensor of real-valued representations. Furthermore, predicates are interpreted as functions or tensor operations that project onto a value within the interval [0, 1]. Notably, these functions are typically implemented using neural networks. The grounding mechanism, denoted by $\mathcal{G}$, associates a tensor of real numbers, where each real number falls within the interval [0, 1].

3. Our methodology

Let $X = \{x_i, p_i\}_{i=1}^{N}$ denote the collection of labeled data, where each x_i is an input tweet text and p_i is the corresponding target, with N being the number of instances. Each sentence–target pair (x_i, p_i) has an associated stance label y_i. Stance detection aims to predict the stance label y using a model trained on this labeled data.

As illustrated in Fig. 1, the FOLAR framework comprises three components: KE, LTN, and MDF. The specific details of each component are subsequently delineated in Sections 3.1, 3.2, and 3.3, respectively.

3.1. Knowledge elicitation

To enhance LLMs' reasoning ability, we first propose a CoT prompting approach using FOLs. The FOL aids in organizing reasoning and visualizing decisions for stance detection, as exemplified in Fig. 2. First, a task is constructed for the LLM. Next, we design a prompt template given sentence "*The narrow libertarian types never stop to think what it costs to NOT have intercity passenger rail. And one of the costs is airport delays, and the cost of adding another runway, etc. And rail is safer per mile traveled - much safer*" and target "public rail". We give FOL as an example for the LLM:

$$\forall x((\neg R(x) \rightarrow (A(x) \vee C(x))) \vee (R(x) \rightarrow S(x))) \rightarrow Support \tag{1}$$

where "R(x): x is rail. A(x): x causes airport delays. C(x): x costs more. S(x): x is safer". These can be formulated based on human expert knowledge or selected by querying the LLM before constructing the chain. Finally, given a new input, the LLM can automatically generate FOLs. To validate the efficacy of the KE method, case studies are presented in experiments (see Table 5).

3.2. Logic tensor network for stance detection

To effectively leverage annotated data for learning stance detection, we further construct LTNs to model the acquired FOLs. Specifically, we first give the definition of LTN in Section 3.2.1. Details of the neural network structure of LTN are described in Section 3.2.2.

3.2.1. LTN definition

Important terms are defined for the formulation of LTN for stance detection:

- **Domains**:
  texts, representing the examples from dataset.
  targets, representing the target related to texts.
  labels, representing the class labels.
- **Variables**:
  x_+, $x_\circ$ and x_- denote the text with "favor", "neutral" and "against" stance polarity, respectively.
  p denotes the given target.
  $\mathrm{D}(x_+) = \mathrm{D}(x_\circ) = \mathrm{D}(x_-) = \mathrm{D}(\mathrm{x}) =$ *texts*.
- **Constants**:
  l_+, $l_\circ$ and l_-, denoting the stance polarity classes/labels for "favor", "neutral" and "against", respectively.
  $\mathrm{D}(l_+) = \mathrm{D}(l_\circ) = \mathrm{D}(l_-) =$ *labels*.
- **Axioms**: Given that the softmax operator in DNNs yields a continuous real-valued output in [0,1], we adopt soft logic (Bianchi & Hitzler, 2019). Rather than Boolean logic results of 0 or 1, this logic generates a continuous value in [0,1] as output. Leveraging soft logic enables representing ambiguity, which is neither entirely correct nor entirely incorrect.
  Concurrently, followed by soft logic (Badreddine et al., 2022), we extend the logical operators negation "¬", conjunction "∧", disjunction "∨", logical implication "→" as:

$$\neg U = 1 - U \tag{2}$$

$$U_1 \wedge U_2 = U_1 \cdot U_2 \tag{3}$$

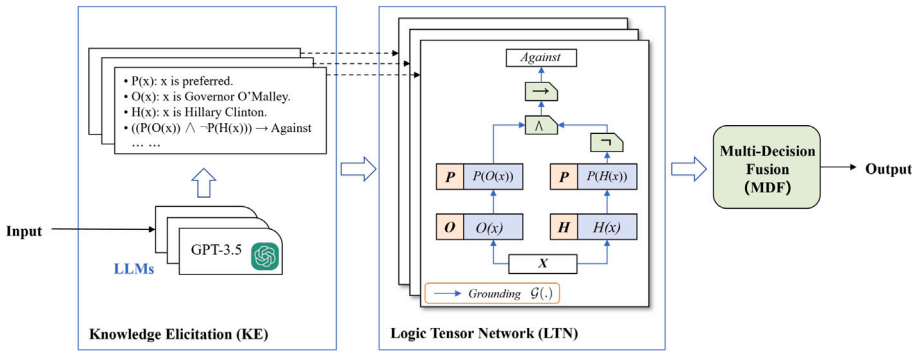


Fig. 1. Overall framework of our method. First, the input is fed through the KE, which leverages the capabilities of LLMs to acquire FOLs. Then, the acquired FOLs are utilized to construct LTNs. Third, the MDF module aggregates multiple LTNs, enabling prediction.

Your task is to analyze the attitude of the given sentence towards the given target and formulate a response using first-order logic. Conclude with an attitude (Support, Opposed, Neutral) statement at the end.

Input:
Sentence: "The narrow libertarian types never stop to think what it costs to NOT have intercity passenger rail. And one of the costs is airport delays, and the cost of adding another runway, etc. And rail is safer per mile traveled - much safer"
Given target: "public rail"

Output:
First-order logic:
R(x): x is rail.
A(x): x causes airport delays.
C(x): x costs more.
S(x): x is safer.
Support: Support the given target "public rail".
∀x ((¬R(x) → (A(x) ∨ C(x)) ∨ (R(x) → S(x))) → Support

Input: x

Output:

Fig. 2. An illustration of knowledge elicitation.

$$U_1 \vee U_2 = U_1 + U_2 - U_1 \cdot U_2 \tag{4}$$
$$U_1 \rightarrow U_2 = 1 - U_1 + U_1 \cdot U_2 \tag{5}$$

- **Grounding**:
  $\mathcal{G}(l)$ is the one-hot vector where $\mathcal{G}(l_+) = [1,0,0]$, $\mathcal{G}(l_n) = [0,1,0]$ and $\mathcal{G}(l_-) = [0,0,1]$
  $\mathcal{G}(x)$ is a word matrix of x.
  $\mathcal{G}(P|\theta)$: (x, p), $l \mapsto \text{softmax}(\text{LTN}_\theta(x, p))$, where the LTN has three output neurons corresponding to the stance polarity "favor", "neutral" and "against", and each neuron gives the probability corresponding to the class l.

3.2.2. Neural network structure of logic tensor network

Following the LTN definition, we aim to construct $\mathcal{G}(P|\theta)$: (x, p), $l \mapsto \text{softmax}(\text{LTN}_\theta(x, p))$ by DNN structure, thereby enabling the use of labeled samples to backpropagate and update the network parameters. Specifically, following FOL form, $\mathcal{G}(P|\theta)$ can be decomposed into each *grounding*, which is constructed by the neural network layer.

In this work, we utilize pretrained Bert (Devlin, Chang, Lee, & Toutanova, 2019) to obtain groundings for diverse input texts. To clearly explain the model structure, we define a toy decision process as follows: "$(a \wedge b) \rightarrow l$". Fig. 3 provides the framework. To achieve the first layer *grounding* $\mathcal{G}(a)$, $\mathcal{G}(b)$ and $\mathcal{G}(l)$, we first let the input texts $\{a, b, l\}$ pass through the pretrained Bert to obtain embedded word vectors e_a, e_b, e_l. Each grounding $\mathcal{G}(.)$ is then represented through a transformation performed by a tensor operation on e_a, e_b, e_l. Inspired by Huang et al. (2022), we employ a fully connected layer to directly acquire this representation. For instance, the vector representation of $\mathcal{G}(a)$ is obtained by feeding e_a into a fully connected layer. Subsequently, the complete computational graph can be constructed based on the soft logic operations in Section 3.2.1. Finally, the representation of this expression, denoted as q can be computed as follows:

$$q := 1 - (\mathcal{G}(a) \cdot \mathcal{G}(b)) + (\mathcal{G}(a) \cdot \mathcal{G}(b)) \cdot \mathcal{G}(l) \tag{6}$$

3.3. Multi-decision fusion

Given that different LLMs vary in their training data and methods, their reasoning processes for stance detection also differ. To improve the fusion of decisions from multiple models, we propose the MDF approach. Specifically, inspired by Zhang, Li et al. (2020) on multi-knowledge fusion, MDF adopts an attention-based dynamic routing strategy. Formally, we first initialize the decision representation vector r, viewed as the integrated knowledge from all decisions. Next, we deploy the KE method for multiple different LLMs to elicit FOL, then construct an LTN for each based on the obtained FOLs. In this paper, we employed three LLMs to elicit FOLs, namely GPT-3.5, LLama-2, and Claude-2. The vector representations q from each LTN are concatenated

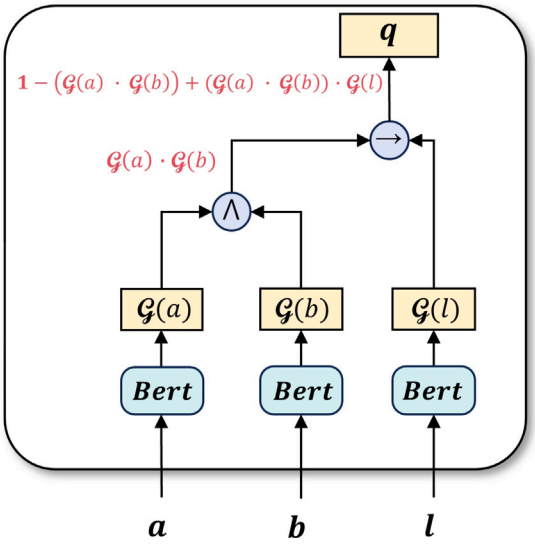


Fig. 3. An illustration of toy FOL "$(a \wedge b) \rightarrow l$".

to form H, which are treated as features to be fused. Formally, the coupling coefficient c can be calculated as:

$$c^1 = r^1 (H^1)^{\mathrm{T}} \tag{7}$$

Then, the next iterative can be calculated by:

$$r^2 = c^1 H^1 \tag{8}$$

where the dimension of the new query r^2 is the same as r^1. Then, the representation of the next iteration can be computed by:

$$H^2 = \lambda \, \mathcal{LN}(\sigma(r^2)) + H^1 \tag{9}$$

where $\mathcal{LN}$ performs the standard layer normalization. σ denotes the *Sigmoid* function. After t iterations, the final representation r^t can be calculated as:

$$r^t = H^t(S \sum_i c^t) \tag{10}$$

where S denotes softmax operator.

After obtaining the r^t, which is then passed through a fully connected layer with a softmax activation function. This process yields the final probability distribution as:

$$\hat{y} = softmax(W_c r^t + b_c) \tag{11}$$

where W_c and b_c are trainable parameters. The training objective of the model is to minimize the cross-entropy loss between the predicted probability distribution and the true labels, as follows:

$$Loss = -\sum_{i=1}^{N} \sum_{j=1}^{C} y_{ij} \log \hat{y}_{ij} \tag{12}$$

where N is the number of instances in the training set, C is the number of label classes.

4. Experiments

In this section, we introduce the datasets, implementation details, evaluation metrics employed in the experiments and the baseline methods used during the experiments.

Table 1
Statistics of SEM16 and ISD datasets.

	Target	Favor	Against	Neutral
SEM16	DT	148	299	260
	HC	163	565	256
	LA	167	544	222
	FM	268	511	170
ISD	DT_I	875	1096	1134
	JB_I	1046	912	525

Table 2
Statistics of VAST dataset.

	Train	Dev	Test
#Examples	13,477	2062	3006
#Unique Comments	1845	682	786
#Zero-shot Topics	4003	383	600
#few-shot Topics	638	114	159

4.1. Experimental data

To evaluate the performance of FOLAR, we conduct comprehensive experiments on the SemEval-2016 Task 6 benchmark (SEM16) (Mohammad, Kiritchenko, Sobhani, Zhu, & Cherry, 2016), the VAST corpus (Allaway & Mckeown, 2020), and the ISD dataset (Huang et al., 2023).

SEM16. The SEM16 dataset includes 6 predefined targets spanning multiple domains, namely Feminist Movement (FM), Hillary Clinton (HC), Donald Trump (DT), Legalization of Abortion (LA), Atheism (Ath), and Climate Change (CC). Each instance is categorized as "Favor, Against or Neutral". Following Zhang, Yang et al. (2020), we exclude DT, Ath, and CC due to data quality concerns. Additionally, following Liang, Chen et al. (2022), we use one of the remaining targets for zero-shot testing and training on the others. 15% of the training data is randomly selected as the development set for hyperparameter tuning. To differentiate in-target and zero-shot settings, we abbreviate the settings as (HC, FM, LA, and DT).

VAST. VAST comprises diverse targets, with each example containing a sentence, target, and stance polarity of either "Pro, Con, or Neutral". Adhering to Li, Zhao and Caragea (2023), we evaluate our model's performance on zero-shot topics under 100% and 10% training data conditions, respectively.

ISD. The ISD dataset includes a total of 6,027 labeled tweets, divided between Donald Trump (DT_I) and Joe Biden (JB_I) targets. Every sentence in ISD incorporates at least one hashtag representing an attitude-bearing topic, with an average of 2.14 hashtags per sentence. The dataset is split into training and testing sets with an 80/20 ratio. The statistics of the datasets are presented in Tables 1 and 2.

4.2. Implementation details

In the LTN model, predicates are represented by phrases or short sentences. For grounding these predicates, we use a pretrained Bert (Devlin et al., 2019) as a mapper to project different predicate phrases into a unified representational space. This design ensures that all predicates share the same parameters of the pretrained Bert model rather than initializing new random weights for each predicate or sentence. Consequently, this unified space enables consistency across predicates without requiring a separate grounding for each new instance. LTN was trained using the AdamW optimizer with a batch size of 16, a maximum of 30 epochs, and a warm-up ratio of 0.2. The learning rate was set to 1e–5, and a weight decay of 1e–3 was employed. For the LLMs, we use the gpt-3.5-turbo-030,[1] Claude-2-100k,[2] and LLama 2-70b[3] models,

[1] https://platform.openai.com/docs/models/gpt-3-5.
[2] https://www.anthropic.com/news/claude-2.
[3] https://huggingface.co/meta-llama/Llama-2-70b-chat-hf.

with the temperature set to zero to ensure reproducibility. The model is trained with Nvidia A100 GPUs and Pytorch.

4.3. Evaluation metrics

The SEM16, ISD, and VAST datasets comprise three-class classification tasks, where each sample is annotated with one of three labels: "favor", "against", or "neutral (none)". For SEM16 and ISD, we adopted the experimental setup as (Huang et al., 2023) and reported the average F1-score.

$$F_{avg} = \frac{(F_{favor} + F_{against})}{2} \tag{13}$$

In the case of VAST, we followed the setup described in Li, Zhao et al. (2023) and Liang, Zhu et al. (2022) and reported the average F-score.

$$F_{avg} = \frac{(F_{favor} + F_{against} + F_{None})}{3} \tag{14}$$

4.4. Baseline methods

To evaluate the performance of our FOLAR across diverse datasets, we employed the following models as the baseline for comparison:

Statistics-based methods:

- **Bicond** (Augenstein, Rocktaeschel, Vlachos, & Bontcheva, 2016) utilizes two BiLSTM models to individually encode the underlying sentences and their associated targets.
- **SEKT** (Zhang, Yang et al., 2020) leverages external knowledge, particularly semantic and emotion lexicons, to facilitate knowledge transfer across diverse targets for cross-target stance detection.
- **CrossNet** (Du, Xu, He, & Gui, 2017a) encodes text and topic using identical bidirectional architectures as BiLSTM and incorporates a target-specific attention layer prior to classification for cross-target stance detection.
- **TPDG** (Liang et al., 2021) presents a novel technique that autonomously distinguishes and calibrates the target-dependent and target-independent roles of a term regarding a specific target within stance expressions.

PLM-based methods:

- **JoinCL** (Liang, Zhu et al., 2022) comprises stance contrastive learning and target-aware prototypical graph contrastive learning for generalizing target-dependent stance features to unseen targets.
- **PT-HCL** (Liang, Chen et al., 2022) proposes a contrastive learning approach utilizing both semantic and sentiment knowledge to improve cross-domain transferability of features.
- **TarBK** (Zhu et al., 2022) incorporates specific background knowledge from Wikipedia to reduce the semantic gap between known and unseen targets, enhancing generalization and reasoning.
- **SDAgu** (Zhang, Wu, Zhang, & Feng, 2023) presents a self-supervised data augmentation method leveraging coreference resolution to mitigate limited labeled data and implicit stance expression.
- **KPT** (Shin, Razeghi, IV, Wallace, & Singh, 2020) leverages external lexicals to initialize the verbalizer component, which is embedded within the prompt framework, to facilitate the integration of domain-specific knowledge.
- **KEprompt** (Huang et al., 2023) employs a topic model to acquire hashtag representations and then performs prompt-tuning methods for stance detection.

Table 3
Performance comparison of in-target stance detection. The best scores are in bold.

	Methods	SemEval-2016			ISD	
		FM	LA	HC	DT_I	JB_I
sta.	TPDG[a]	67.3	74.7	73.4	64.2	60.0
Bert	BERT-FT[a]	62.3	62.4	67.0	69.1	65.6
	KPT[a]	63.3	63.5	71.3	69.4	66.4
	KEprompt[a]	**72.1**	69.1	74.4	70.5	67.4
LLM	Ts-CoT	67.2	62.5	83.2	64.7	68.5
	KASD[b]	70.4	63.3	77.6	71.1	70.9
	MPPT*	69.6	68.1	82.1	71.3	70.6
	COLA[c]	69.1	71.0	75.9	–	–
	FOLAR	70.6	**72.7***	**84.5***	**73.3***	**71.5***
	FOL-CoT	70.1	68.4	83.6	67.6	70.2
	w/o MDF	69.1	71.3	85.5	71.5	70.9

* Refers to a p-value < 0.05.
[a] The Results are retrieved (Huang et al., 2023).
[b] From Li, Liang et al. (2023).
[c] From Lan et al. (2023).

LLM-based methods:

- **Ts-CoT** (Zhang, Fu, Ding et al., 2023) develops a CoT method for LLMs to perform stance detection. Due to the evolution of LLM model versions, the foundation model is upgraded to the GPT-3.5 baseline.
- **KASD** (Li, Liang et al., 2023) utilizes LLMs to generate background knowledge, that enhances the stance detection performance.
- **MPPT** (Ding et al., 2024) enhances stance detection effectiveness by constructing prompt templates from diverse analytical perspectives of LLMs and leveraging corresponding background knowledge.
- **COLA** (Lan, Gao, Jin, & Li, 2023) COLA enhances the effectiveness of LLM-based stance detection by incorporating multiple LLMs and proposing a model fusion strategy.
- **FOL-CoT** is a variant of our approach, which directly leverages the CoT method to derive prediction results from LLMs.

4.5. Result

To evaluate the stability of the model, following Liang, Chen et al. (2022) and Liang, Zhu et al. (2022), we ran the method three times and reported the F_{avg}. The main experimental results are reported in Tables 3 and 4. Our method achieves the best performance in most of the experimental setups, verifying the effectiveness of our proposed approach for stance detection. Furthermore, compared to previous models, our method shows statistically significant improvements (p-value < 0.05) in most evaluation metrics.

4.5.1. In-target setup

Table 3 presents in-target stance detection results compared against strong baselines. Several key conclusions can be drawn from the results. First, pretrained models substantially outperform statistical methods in most setups. As an example, BERT-FT improves F1 score by 5.3% over the top statistical model TPDG on ISD, demonstrating the efficacy of pretrained models. Second, prompt-tuning methods consistently improve over fine-tuned PLMs, with KEprompt surpassing fine-tuned BERT-FT by 6.7% on LA, better leveraging PLMs. KEprompt achieves improvements on most tasks, which gains stem from its automatic verbalization and integration of background knowledge. Third, by using the simple CoT method (e.g., Ts-CoT) without needing backpropagation learning, LLMs can directly provide significant performance gains,

Table 4
Performance comparison of zero-shot stance detection. The best scores are in bold.

	Model	SEM16				VAST (100%)			VAST (10%)		
		HC	FM	LA	DT	Pro	Con	All	Pro	Con	All
Sta.	Bicond	32.7	40.6	34.4	30.5	44.6	47.4	42.8	29.8	40.1	34.8
	CrossNet	38.3	41.7	38.5	35.6	46.2	43.4	43.4	37.3	32.9	36.2
	SEKT	50.1	44.2	44.6	46.8	50.4	44.2	41.8	–	–	–
	TPDG	50.9	53.6	46.5	47.3	53.7	49.6	51.9	–	–	–
Bert	BERT-FT	50.1	42.1	44.8	41.0	56.8	67.6	71.0	50.7	56.3	64.9
	JointCL	54.4	54.0	50.0	50.5	64.9	63.2	71.2	53.8	57.1	65.5
	TarBK	55.1	53.8	48.7	50.8	65.7	63.9	73.6	–	–	–
	PT-HCL	54.5	54.6	50.9	50.1	61.7	63.5	71.6	–	–	–
LLM	SDAgu	–	–	–	–	60.1	65.2	71.3	–	–	–
	Ts-CoT	83.2	67.2	62.5	61.5	68.7	63.4	73.4	56.5	66.0	70.3
	KASD	80.3	70.4	62.7	**71.2**	–	–	76.8	–	–	–
	MPPT	83.6	70.9	70.1	70.7	69.2	77.6	74.4	67.9	68.2	74.0
	COLA	75.9	69.1	71.0	**71.2**	73.4	77.2	73.4	–	–	–
	FOLAR	**85.3***	**73.7***	**71.9***	70.9	**74.0***	**78.6***	**81.0***	**72.1***	**76.8***	**80.2***
	FOL-CoT	83.6	70.1	68.4	65.9	77.7	72.8	79.8	77.7	72.8	79.8
	wo/MDF	74.1	71.8	71.2	69.7	75.1	73.4	75.2	73.7	72.3	74.9

* Refers to a p-value < 0.05.

outperforming comparative algorithms on most datasets. Notably, the approach that leverages LLMs to generate knowledge and combines it with fine-tuning training methods (MPPT and KASD) has achieved the current highest F1 score. This is attributed to the effective integration of LLM capabilities and annotated labels. Finally, the proposed FOLAR method demonstrates superior performance compared to all baseline models across most tasks. For example, FOLAR boosts F1 scores by 1.7% on LA and 2.0% on DT_I over the strongest competitor.

4.5.2. Zero-shot setup

Collecting large annotated datasets is time-intensive and laborious. Therefore, we evaluate our method in a zero-shot setup, inferring stances towards unseen targets using data from a source target. The result is shown in Table 4. Specifically, we first observe that most statistics-based and fine-tuning based methods perform poorly under zero-shot settings, owing to their reliance on seen targets and texts. Directly introducing pretrained Bert (BERT-FT) did not improve performance, likely because PLMs cannot capture implicit stance expressions absent from the text, such as background knowledge and topics related to the target. After incorporating background knowledge (TarBK), classification accuracy significantly increased. This validates the efficacy of incorporating background knowledge. Furthermore, we observed that contrastive learning methods outperformed fine-tuning models. This is because contrastive learning can effectively capture domain-invariant representations, thereby improving model transferability across domains. Moreover, the application of LLMs has led to a significant performance boost in stance detection under zero-shot settings, with F1 scores surpassing those of statistics-based and fine-tuning methods. Notably, our FOLAR approach achieves significant performance gains across two datasets and three experimental settings. This thoroughly validates the effectiveness of our proposed method. Specifically, the results demonstrate more pronounced improvements under the fine-grained zero-shot setting, advancing performance by 4.2% on VAST (100%) and 6.2% on VAST (10%) over the best competitor.

In summary, FOLAR significantly outperforms current approaches on most tasks under the in-target and zero-shot settings. The superiority of our method stems from three key aspects: (1) Incorporating FOLs enables effective reasoning of the decision process for accurate prediction; (2) Constructing the prediction model with LTN can efficiently exploit available labeled samples to enhance reasoning capacity; (3) MDF can effectively fuse the inferential capabilities of different LLMs. Notably, we further present experimental results for FOL-CoT, which elicits knowledge directly from the FOL extracted from models without relying on labeled data for backpropagation learning. The results demonstrate that FOL-CoT achieves substantial performance gains by providing LLMs with FOL-based reasoning chains in a one-shot manner. In particular, compared to the Ts-CoT approach, the introduction of FOL leads to significant improvements, validating that the knowledge elicitation based on CoT imparts LLMs with the capacity for deeper reasoning over text.

4.6. Ablation study

In this study, we conducted a series of ablation experiments to evaluate the individual contributions of each key component within the FOLAR model. These components include the FOL, LTN, and MDF. Each experiment was designed to isolate the effect of one component by removing it from the model, thereby helping us understand its specific impact on the overall performance.

Ablation of the FOL Component. In the first ablation scenario, labeled as "w/o FOL", we modified the FOLAR model to exclude the FOL component. Without the FOL, the architecture of the network closely resembles that of the Ts-CoT model, which does not incorporate logic-driven feature integration. The results from this ablation test, as shown in Fig. 4, indicate a significant drop in classification accuracy. This decline underscores the critical role of FOL in enhancing the model's reasoning capabilities, which facilitates a structured and logical flow in predictions. By integrating FOL, the model leverages logical deductions to improve the accuracy and robustness of its outcomes, which is particularly advantageous in complex decision-making tasks.

Ablation of the LTN Component. The second ablation test, denoted as "w/o LTN", involved removing the LTN module from FOLAR. In this setup, the predictions of the model are generated directly from the LLM, without the intermediate logical processing typically provided by the LTN. The absence of the LTN resulted in a decrease in performance. This decline can be attributed to the model's reduced ability to learn from canonical examples and to process these using a structured logical framework. The LTN plays a vital role in translating complex logical constructs into trainable features for the LLM, enhancing the model's ability to generalize across different types of data and scenarios.

Ablation of the MDF Component. Finally, the "w/o MDF" involved utilizing only a single LLM, specifically GPT-3.5, rather than employing a diverse set of models as in the full FOLAR configuration. The results, illustrated in Fig. 4, show that performance suffers when the MDF is not used. This finding supports the hypothesis that diversity in the underlying models can mitigate biases and improve the robustness of the predictions. The MDF component enriches the model's training and inference stages by integrating diverse perspectives and methodologies from multiple LLMs, thus reducing the dependency on

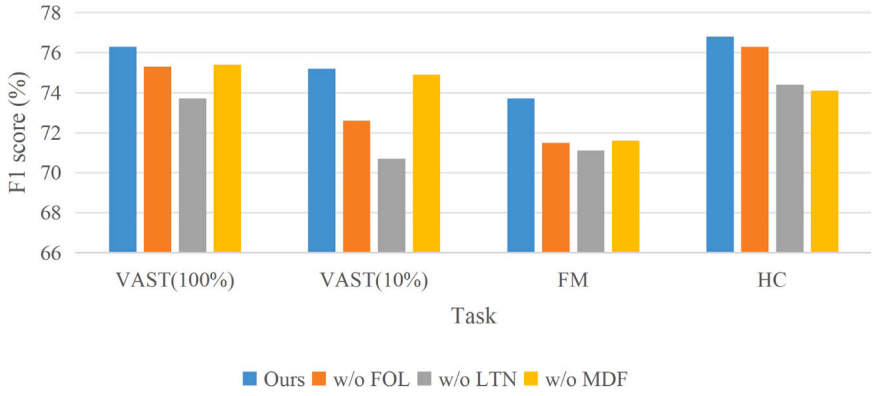


Fig. 4. Experimental results of ablation study.

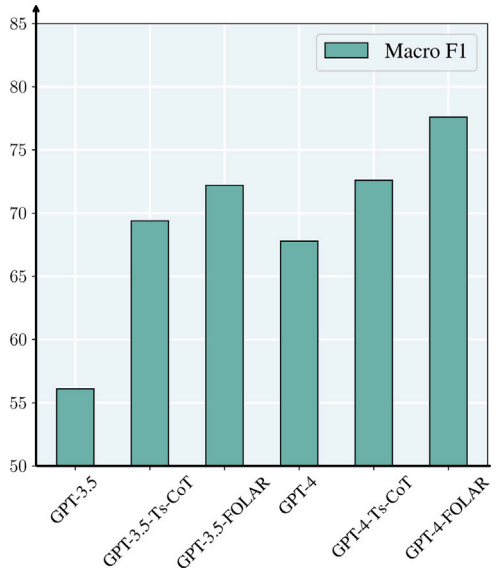


Fig. 5. Experimental results of different GPT models.

any single model's biases or limitations. Additionally, in Fig. 5, we present the average performance on four targets of our method using GPT-4 and GPT-3.5 on the SEM16 dataset in a zero-shot setup. Notably, the results demonstrate that our approach is not limited to a specific large model and can be further improved with more advanced models, such as GPT-4, leading to more significant performance gains.

Summary of Findings. The comprehensive results from all ablation studies clearly demonstrate that each component of the FOLAR model (FOL, LTN, and MDF) significantly contributes to its overall effectiveness. The integration of all these components leads to optimal performance across various experimental datasets, highlighting the synergistic benefits of combining these innovative features. As anticipated, the full FOLAR model, which leverages the combined strengths of FOL, LTN, and MDF, achieves superior performance, validating the design and implementation of these components in enhancing the model's predictive capabilities.

4.7. Case study

Table 5 presents three instances where our model, FOLAR, successfully predicted the correct categories using FOL expressions, whereas the baseline model failed.

These examples underscore the effectiveness of incorporating FOL in the decision-making process, demonstrating that LLMs can be enhanced to not only generate, but meaningfully interpret logical constructs. The ability of FOLAR to handle complex logical expressions, as evidenced by these examples, shows that it effectively interprets special symbols in FOL. This capability is crucial for encoding logical relationships and reasoning paths, which are integral to improving prediction accuracy and model robustness against simpler models. Moreover, integrating a CoT approach enables deeper reasoning over text. This method encourages LLMs to articulate their reasoning process step-by-step, aligning the model's output with human-like logical deductions. This structured reasoning leads to better performance, as LLMs can efficiently process and analyze the information before making a decision. In conclusion, the examples in Table 5 highlight the significant benefits of embedding structured logical reasoning and decision rationale in LLMs. FOLAR's performance advantage over the baseline model illustrates the practical value of our approach in complex decision-making tasks.

Table 5
Examples of FOL expression generated by LLM.

Input	FOL
Text: Bikes lanes are one of the dumbest ideas Bloomberg has ever had. Large areas of Manhattan have now been reduced to single or double lanes where a taxi or two picking up or discharging a passenger stop traffic completely. Moronic. **Target**: bike lane	• B(x): x is a bike lane. • C(x): x causes traffic congestion. • D(x): x is a dumb idea. • O(x): x is Against. • ∀ x ((B(x) → C(x))) ∨ (B(x) → O(x))) → Against
Text: I know that in today's atmosphere, I would never be a school teacher, scout leader, or Sunday school teacher without a Digital Video Record of my every encounter with children. I suspect some police officers feel the same qualms dealing with the public. **Target**: camera	• C(x): x is a camera. • N(x): x is necessary. • S(x): x feels the same way. • T(x): x is a school teacher, scout leader, or Sunday school teacher. • P(x): x is a police officer. • D(x, y): x has a digital video record of y. • ∀x ((T(x) ∨ P(x)) → N(D(x, children))) ∧ (P(x) → S(x, public)) → Support
Text: Although I certainly have disagreements, after reading about @GovernorOMalley I much rather have him than @HillaryClinton. **Target**: Hillary Clinton	• P(x): x is preferred. • O(x): x is Governor O'Malley. • H(x): x is Hillary Clinton. • ∀x ((P(O(x)) ∧ ¬P(H(x))) → Against

4.8. Error analysis

To better comprehend the limitations of FOLAR, we further analyze the errors committed by FOLAR. Specifically, we randomly select 50 instances from the VAST dataset that are incorrectly predicted by FOLAR. The FOLs are shown in Table 6. Our analysis unveils several underlying sources for the classification errors, which can be categorized as follows.

Table 6
Error examples of FOL expression generated by LLM.

Input	FOL
Text: It is simple. We need a national biometric id system that allows humans to show up at the polls and present themselves to vote. The idea that only the members of the 'master race' can vote will only lead to trouble. **Target**: constitutional right	• M(x): x mentions the term "master race". • N(x): x contains negative terms related to the target. • T(x): x has the target "constitutional right". • O(x): x is Against. • ∀ x ((T(x) ∧ (M(x) ∨ N(x))) → O(x))
Text: Airports and the roads on east nor west coast cannot handle the present volume adequately as is. I did ride the vast trains in Europe, Japan and China and found them very comfortable and providing much better connections and more efficient. **Target**: Traffic	• T(x): x is about the target "traffic". • P(x): x contains positive terms or sentiments. • O(x): x is Favor. • ∀ x (T(x) ∧ P(x) → O(x))

First, FOLAR struggles to classify sentences with implicit viewpoints or requiring deeper understanding. For example, for the sentence "*It's simple. We need a national biometric id system that allows humans to show up at the polls and present themselves to vote. The idea that only the members of the 'master race' can vote will only lead to trouble.*$_{[favor]}$" with a target "constitutional right", FOLAR tends to predict an incorrect *against* stance. This is because the FOLAR model cannot comprehending the sarcastic semantics with 'master race', which is not acquirable from the KE method.

The second error category is caused by over-inferencing intrinsic to the LLMs themselves. For example, FOLAR cannot correctly predict the stance for the sentence "*Airports and the roads on east nor west coast cannot handle the present volume adequately as is. I did ride the vast trains in Europe, Japan and China and found them very comfortable and providing much better connections and more efficient.*$_{[against]}$" with a target "traffic". This could be attributed to the opposing stance polarities present in the first and second clauses of the sentence, alongside the LLM associating the "comfortable trains" in the latter clause with the target "traffic" as well.

4.9. Manual evaluation

Due to the novel proposal of a CoT approach representing decision rationale as FOL and the significant performance boost, existing methods lack accurate assessment of FOL validity. Therefore, we manually evaluate the quality of the generated FOL expressions.

Specifically, we invited three NLP researchers with master's degrees to rank these explanations based on three criteria: (1) The generated explanations contain salient, significant information without omitting any major viewpoints conducive to stance detection. (2) The generated explanations do not include any redundant, repetitive or irrelevant information to the input and stance detection. (3) The generated explanations contain no contradictory information to the input and stance detection.

We randomly sampled 100 instances from the test set and computed kappa statistics (McHugh, 2012) to measure agreement. The results show that the kappa scores for all three test objectives exceeded 70%, with an average score of 76%, validating the high quality of the FOL-based decision rationale representation.

5. Conclusion

This paper proposed the FOLAR framework, a novel approach that harnesses the power of first-order logic and chain-of-thought reasoning for stance detection. Through the knowledge elicitation module, FOLAR utilizes large language models to derive logical inference rules that articulate the rationale behind predictions. These rules are then integrated into logic tensor networks, which are trained on annotated datasets to enhance accuracy and interpretability. Then the multi-decision fusion component of FOLAR synthesizes diverse reasoning paths from multiple models, thereby enhancing the robustness and reliability of the stance detection. Our extensive experiment on three benchmark datasets has demonstrated that FOLAR not only sets a new benchmark in stance detection performance but also provides deeper interpretative insights into the logic of stance reasoning. Looking forward, potential future work could involve exploring the extraction of more complex forms of logic and expanding the application of the FOLAR framework to additional contentious topics or domains. Such expansions could further validate and refine the framework's capabilities. Additionally, investigating the integration of newer, more dynamic forms of logic and enhancing the decision fusion mechanism could lead to even more robust stance detection systems.

CRediT authorship contribution statement

Genan Dai: Methodology, Conceptualization. **Jiayu Liao:** Resources, Methodology. **Sicheng Zhao:** Formal analysis, Data curation, Conceptualization. **Xianghua Fu:** Resources, Investigation. **Xiaojiang Peng:** Formal analysis. **Hu Huang:** Project administration, Methodology. **Bowen Zhang:** Methodology, Investigation, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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